

Torsade v1 — ECG Noise and Lead-Failure Stress-Test Corpus: Generation Report and Full Record Roster

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Date: 2026-07-13 · **Repository:** <https://github.com/depolarised/torsade> (tag v1.0.0) · **Sources (every value below traces to these):** `recipes/corpus.yaml`, `recipes/source_ids/*.csv`, `manifest.json` (all committed), regenerated in-memory from `torsade.corpus + torsade.recipes` (`master_seed = 20260713`). Generator: `scripts/reports/generate_corpus_report.py`.

Scope. Torsade is a *stress-test set*, not a *clinically representative cohort*. It deliberately over-samples failure; do not use it to estimate real-world prevalence or deployment accuracy.

1. Objectives

Torsade is a deterministic generator and reproducible corpus **definition** for stress-testing ECG signal-quality gates (the internal `signalguard`) and noise detectors (`noiseguard`). It does **not** redistribute derived signals: it ships code, recipes, labels, manifest, and provenance so a user regenerates the corpus **bit-exactly** from their own open PhysioNet copies.

This report documents the generation algorithm and enumerates **all 67 stress records** with their corruption truth and expected behaviour. Records resolve against a local PhysioNet mirror; the concrete PTB-XL parent ids shown here are those in `recipes/source_ids/` (reproducible, seeded selection).

2. Corpus composition

Group	n	Source	Corruption
Naturally poor	15	PTB-XL technical-validation quality flags	none (inherently noisy)

Group	n	Source	Corruption
Real-noise pairs	30	clean PTB-XL + NSTDB em/ma/bw	SNR ladder {-6, +0, +6, +12, +18} dB
Engineering extremes	22	clean PTB-XL parents (+ MACECGDB motion)	single-lead (12) + multi-lead (10)
Total	67		11 labelled data-integrity failures

Each record carries three label layers — **clinical parent** (rhythm / Uni-G statements / quality flags), **corruption truth** (electrodes/leads, requested + measured SNR, noise segment, seed, α , amplitude bookkeeping), and **expected behaviour** (per-lead and record-level, mapped to the `signalguard QualityLabel` / `RecordQuality` and `noiseguard discard` vocabularies).

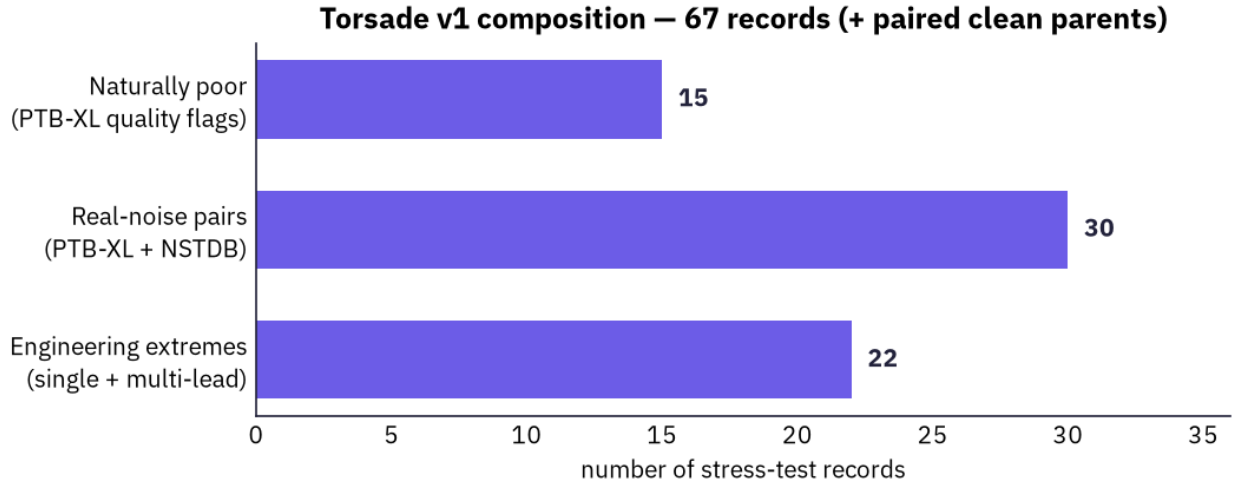


Figure 1: Corpus composition — 67 stress records plus paired clean parents.

3. Sources and licensing

Dataset	Role	Version	License
PTB-XL	clean + naturally-poor parents (500 Hz)	1.0.3	CC-BY-4.0
PTB-XL+	Uni-G statements / rhythm metadata	1.0.1	CC-BY-4.0
MIT-BIH NSTDB	em/ma/bw noise for the SNR ladder	1.0.0	ODC-BY-1.0
MACECGDB	real stand-ing/walking/jumping motion	1.0.0	ODC-BY-1.0

PTB-XL, PTB-XL+, and MACECGDB are read from a local PhysioNet mirror; **only NSTDB** is

fetched by `make download`. **Code** is GPL-3.0-or-later; **manifest, labels, and docs** are CC-BY-4.0.
 No source signals are redistributed — see `ATTRIBUTION.md`.

4. Generation algorithm

4.1 Canonical form. Every parent becomes a (12, T) float64 mV array in canonical lead order [I, II, III, aVR, aVL, aVF, V1-V6] at 500 Hz (NSTDB resampled from 360 Hz by deterministic polyphase resampling). Absent limb leads are derived from I and II.

4.2 Real-noise SNR mixing. For a clean lead x , real noise n , and requested SNR s (dB): $\alpha = \sqrt{P_x / (P_n \cdot 10^{s/10})}$, $y = x + \alpha n$, with P_x , P_n the DC-removed mean-square powers over the contaminated interval only. Both the requested and the *measured* SNR, and the exact NSTDB segment offset, are recorded in the label.

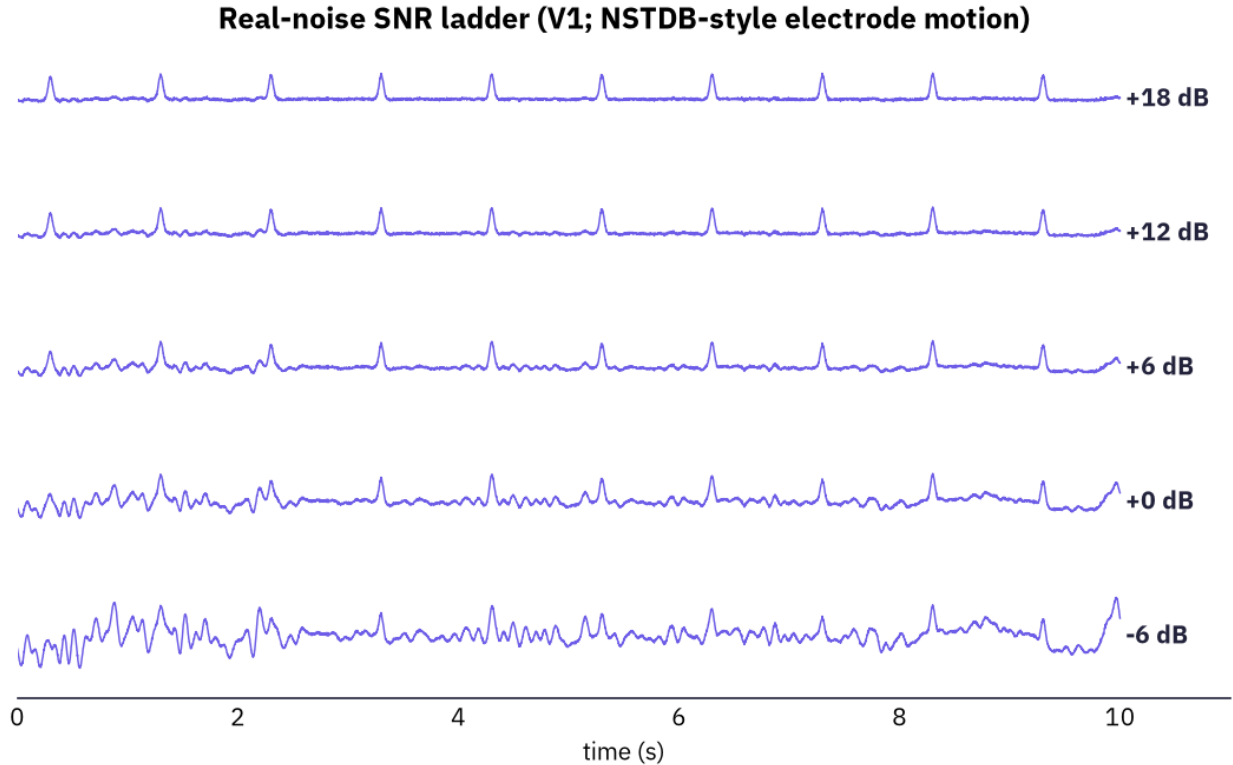


Figure 2: Real-noise SNR ladder (illustrative synthetic parent; NSTDB-style electrode motion).

4.3 Electrode-domain corruption. Limb-electrode potentials are recovered in the RA-as-reference gauge (RA = 0, LA = I, LL = II); chest-electrode potentials as $Ck = Vk + WCT$, $WCT = (RA+LA+LL)/3$. An artefact is added to a chosen electrode; the WCT and all twelve leads are recomputed, and the invalidated derived leads are recorded. (The classical WCT is an idealised electrode model, **not** a torso forward model.)

4.4 Engineering extremes. Deterministic builders for clipping/rail-saturation, flatline (zero), constant-ADC, lead-off, digital-missing (NaN), electrode-motion swings, step-and-exponential-recovery, polarity inversion, and intermittent lead-off. NaN / flat / constant / rail cases are labelled

data_integrity_failure with a type — **not** as noise.

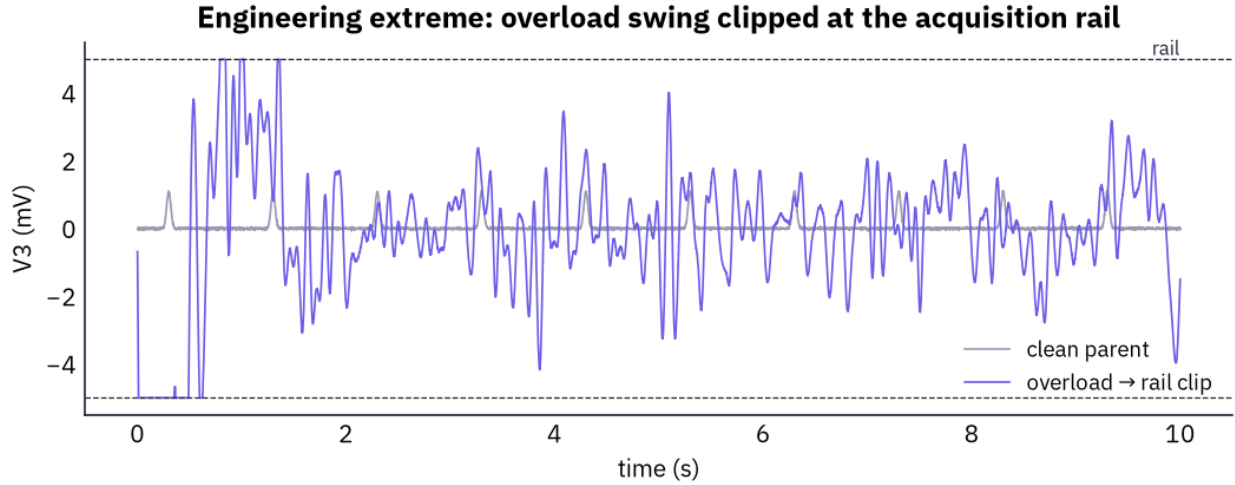


Figure 3: Engineering extreme: an overload swing clipped at the acquisition rail.

5. Full record roster

All 67 records are listed below. **Expected quality** is the record-level **RecordQuality** a correctly-tuned gate should assign; **Discard** is the expected **noiseguard** record-level decision; **Bad leads** is the count of leads the label expects a detector to reject. PTB-XL parents are shown as record stems (the **records500/NNNNN/** directory prefix is omitted).

5.1 Naturally poor (15)

Real PTB-XL records selected by technical-validation quality flags; no clean parent (inherently bad).

Record	PTB-XL source	Quality flag(s)	Expected quality	Discard
torsade_nat_001	10646_hr	electrodes_problems	reject	yes
torsade_nat_002	10652_hr	burst_noise	limited	—
torsade_nat_003	11176_hr	static_noise	limited	—
torsade_nat_004	12587_hr	static_noise	limited	—
torsade_nat_005	13777_hr	static_noise	limited	—
torsade_nat_006	14197_hr	static_noise	limited	—
torsade_nat_007	16149_hr	baseline_drift	reject	yes
torsade_nat_008	16981_hr	static_noise	limited	—

Record	PTB-XL source	Quality flag(s)	Expected quality	Discard
torsade_nat_009	17281_hr	static_noise	limited	—
torsade_nat_010	17293_hr	baseline_drift	limited	—
torsade_nat_011	19487_hr	static_noise	limited	—
torsade_nat_012	20478_hr	static_noise	limited	—
torsade_nat_013	00359_hr	static_noise	reject	yes
torsade_nat_014	06320_hr	baseline_drift	limited	—
torsade_nat_015	06968_hr	static_noise	limited	—

5.2 Real-noise SNR-ladder pairs (30)

Clean PTB-XL parents mixed with NSTDB noise across the ladder $\{-6, +0, +6, +12, +18\}$ dB. Each ships with its untouched clean parent (*_clean).

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
torsade_noise_001	sinus	10236_hr	em @ -6 dB	12	12	reject	yes
torsade_noise_002	afib	10330_hr	ma @ +0 dB	6	0	limited	yes
torsade_noise_003	pac pvc	11089_hr	bw @ +6 dB	6	0	limited	—
torsade_noise_004	bbb conduction	11662_hr	em @ +12 dB	12	0	diagnostic	—
torsade_noise_005	st t	12086_hr	ma @ +18 dB	6	0	diagnostic	—
torsade_noise_006	flutter svt	12407_hr	bw @ -6 dB	6	6	limited	yes
torsade_noise_007	sinus	01306_hr	em @ +0 dB	12	0	limited	yes
torsade_noise_008	afib	13114_hr	ma @ +6 dB	6	0	limited	—
torsade_noise_009	pac pvc	13332_hr	bw @ +12 dB	6	0	diagnostic	—

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
torsade_ noise_ 010	bbb con- duction	13611_hr	em @ +18 dB	12	0	diagnostic	—
torsade_ noise_ 011	st t	14071_hr	ma @ -6 dB	6	6	limited	yes
torsade_ noise_ 012	flutter svt	14211_hr	bw @ +0 dB	6	0	limited	yes
torsade_ noise_ 013	sinus	15473_hr	em @ +6 dB	12	0	limited	—
torsade_ noise_ 014	afib	15535_hr	ma @ +12 dB	6	0	diagnostic	—
torsade_ noise_ 015	pac pvc	15877_hr	bw @ +18 dB	6	0	diagnostic	—
torsade_ noise_ 016	bbb con- duction	16615_hr	em @ -6 dB	12	12	reject	yes
torsade_ noise_ 017	st t	16930_hr	ma @ +0 dB	6	0	limited	yes
torsade_ noise_ 018	flutter svt	17553_hr	bw @ +6 dB	6	0	limited	—
torsade_ noise_ 019	sinus	17727_hr	em @ +12 dB	12	0	diagnostic	—
torsade_ noise_ 020	afib	17770_hr	ma @ +18 dB	6	0	diagnostic	—
torsade_ noise_ 021	pac pvc	17867_hr	bw @ -6 dB	6	6	limited	yes
torsade_ noise_ 022	bbb con- duction	18284_hr	em @ +0 dB	12	0	limited	yes
torsade_ noise_ 023	st t	18508_hr	ma @ +6 dB	6	0	limited	—
torsade_ noise_ 024	flutter svt	19002_hr	bw @ +12 dB	6	0	diagnostic	—
torsade_ noise_ 025	sinus	19238_hr	em @ +18 dB	12	0	diagnostic	—
torsade_ noise_ 026	afib	01979_hr	ma @ -6 dB	6	6	limited	yes

Record	Rhythm	PTB-XL parent	Noise @ SNR	Leads	Bad leads	Expected quality	Discard
torsade_ noise_ 027	pac pvc	20943_hr	bw @ +0 dB	6	0	limited	yes
torsade_ noise_ 028	bbb con- duction	20977_hr	em @ +6 dB	12	0	limited	—
torsade_ noise_ 029	st t	21495_hr	ma @ +12 dB	6	0	diagnostic	—
torsade_ noise_ 030	flutter svt	21820_hr	bw @ +18 dB	6	0	diagnostic	—

5.3 Engineering extremes — single-lead (12)

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
torsade_ eng_001_ emg_v1	02965_hr	swing 6 mV p2p (V1)	—	1	limited	—
torsade_ eng_002_ macecg_ walk_v2	03016_hr	MACECGDB — walk swing 8 mV p2p (V2)	—	1	limited	—
torsade_ eng_003_ intermittent_ v6	03047_hr	intermittent lead-off with recon- nection (V6)	intermittent lead off	1	limited	—
torsade_ eng_004_ step_ii	03365_hr	baseline step 4 mV, tau=1 s (II)	—	0	limited	—
torsade_ eng_005_ macecg_ jump_ overload_ v3	03427_hr	MACECGDB rail jump swing 30 mV p2p (V3) → rail ±5 mV	rail saturation	1	limited	—
torsade_ eng_006_ flatline_ v4	03543_hr	flatline / zero-fill (V4)	flatline zero	1	limited	—
torsade_ eng_007_ constant_ v5	03687_hr	stuck constant 1.2 mV (V5)	constant adc	1	limited	—

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
torsade_ eng_008_ missing_ v6	03730_hr	digital- missing / NaN (V6)	digital missing channel	1	limited	—
torsade_ eng_009_ flatline_ i	03941_hr	flatline / zero-fill (I)	flatline zero	5	limited	—
torsade_ eng_010_ leadoff_ v2	04593_hr	lead-off (V2)	lead off	1	limited	—
torsade_ eng_011_ emg_ burst_v5	00476_hr	swing 5 mV p2p (V5), t in [3,6] s	—	0	limited	—
torsade_ eng_021_ reversal_ i	09509_hr	polarity inversion (I)	—	0	limited	—

5.4 Engineering extremes — multi-lead / mixed (10)

Electrode-domain propagation, precordial coupling, multi-lead saturation, and compound “wild” cases.

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
torsade_ eng_012_ couple_ v1v2	05267_hr	precordial coupling V1, V2 @ +0 dB	—	2	limited	—
torsade_ eng_013_ couple_ v4v5	05298_hr	precordial coupling V4, V5 @ +6 dB	—	0	limited	—
torsade_ eng_014_ triple_ v1v2v3	05887_hr	precordial coupling V1, V2, V3 @ -6 dB	—	3	limited	yes
torsade_ eng_015_ electrode_ la	06175_hr	LA electrode motion 1 mV	—	4	limited	yes
torsade_ eng_016_ electrode_ ra	06526_hr	RA electrode offset 3 mV	—	6	reject	yes

Record	PTB-XL parent	Corruption	Integrity failure	Bad leads	Expected quality	Discard
torsade_ eng_017_ saturate_ v2v3	06789_hr	overload swing 30 mV p2p (V2, V3) → rail ±5 mV	rail saturation	2	limited	—
torsade_ eng_018_ saturate_ v4v5v6	07086_hr	overload swing 30 mV p2p (V4, V5, V6) → rail ±5 mV	rail saturation	3	limited	yes
torsade_ eng_019_ electrode_ ll	08699_hr	LL electrode motion 1.2 mV	—	3	limited	yes
torsade_ eng_020_ compound_ wild	08931_hr	LA electrode motion 0.8 mV ; MACECGDB walk swing 8 mV p2p (V2) ; lead-off (V5)	lead off	5	limited	yes
torsade_ eng_022_ mixed_ integrity	09915_hr	digital- missing / NaN (V1) ; overload swing 25 mV p2p (V6) → rail ±5 mV ; baseline step 3 mV, tau=1.5 s (II)	digital missing channel	2	limited	—

6. Reproducibility and provenance

A single master seed (20260713) plus `provenance.json` (pinned source versions, per-record seed, source-file SHA-256) make the corpus bit-exactly regenerable:

```
make download      # NSTDB only (PTB-XL / PTB-XL+ / MACECGDB from the local mirror)
make select        # resolve PTB-XL parent ids -> recipes/source_ids/ (seeded)
make regenerate    # WFDB records + labels + manifest + provenance -> ./out
```

Determinism is per-record: `SeedSequence(entropy=master_seed, spawn_key=(record_index,))`. The corpus definition leaves source ids unresolved; `resolve_source_ids()` binds the seeded PTB-

XL selection into the specs at generation time.

7. Verification

Confirmed end-to-end on the local PhysioNet mirror (PTB-XL 1.0.3, MACECGDB 1.0.0, NSTDB 1.0.0):

- **Counts:** 67 records (15 / 30 / 22) + 52 paired clean parents.
- **SNR ladder:** max |requested - measured| SNR error **0.0 dB** across all 30 mixes (tol 0.5 dB).
- **Data integrity:** digital-missing (NaN) round-trips through WFDB — the affected channel reads back all-NaN while intact leads stay finite.
- **Determinism:** two independent full regenerations are **byte-identical**.
- **Quality gates:** 57 unit/integration tests pass; **ruff** + **black** clean.

This report is regenerated by `scripts/reports/generate_corpus_report.py`; every figure and table value is computed from the committed corpus definition.